Latest version: https://vuphan314.github.io
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Vu Hoang Nguyen Phan

Curriculum Vitae

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Vu Phan is a Computer Science Ph.D. student of Prof. Moshe Vardi at Rice University, expecting to graduate by 2022 August. He solves computational counting problems with dynamic programming and knowledge compilation. (Vu Phan is a citizen of Vietnam. He was issued an F1 visa and is currently staying in Texas.)

Education

2022/08 **Doctor of Philosophy**, *Computer Science*, Rice University, 3.61/4.00.

2019/12 **Master of Science**, *Computer Science*, Rice University, 3.60/4.00.

2017/08 **Bachelor of Science**, *Computer Science and Math*, Texas Tech University, 4.00/4.00.

Experience

2021 Summer **Graduate Technical Intern**, *Intel Corporation*, Formal Verification.

- Wrote context-free grammar to convert SystemVerilog scoreboards from one formal tool to another.
- Developed scripts to run experiments comparing two formal tools and report results.

Certification

2021 **SVA, Formal, and JasperGold Fundamentals for Designers v19.03**, *Cadence Design Systems*.

Programming Languages

Hardware **SystemVerilog**.

Software **Python, C++; Dafny, Coq**.

Public Projects

2020-2021 **ProCount**, *Projected Counter*, https://github.com/vardigroup/DPMC.

- Implemented a weighted projected #SAT framework with Jeffrey Dudek.

2019-2020 **DPMC**, *Dynamic-Programming Model Counter*, https://github.com/vardigroup/DPMC.

- Implemented a weighted #SAT framework with Jeffrey Dudek.

2019-2020 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, https://github.com/vardigroup/ADDMC.

- Implemented a weighted #SAT solver.

2016-2018 **LED**, *Language of Effective Definitions*, https://vuphan314.github.io/LED.

- Translated the literate programming language LED into SequenceL and LaTeX.

2015-2016 **L**, *Logic*, https://github.com/iensen/LtoASPtranslator.

- Translated the logic programming language L into Answer Set Prolog with Evgenii Balai.

Graduate Courses

2020-2021 **Bioinformatics: Sequence Analysis; Reasoning about Software**.

2019-2020 **Computer Systems Architecture; Statistical Machine Learning**.

2018-2019 **Artificial Intelligence; Compiler Construction, Multi-Core Computing**.

2017-2018 **Logic in Computer Science; Programming Languages**.

Bibliography

Vu Phan is the corresponding author of the following publications (authors sorted by surnames).

Thesis

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: https://scholarship.rice.edu/handle/1911/107761.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: http://jeffreydudek.com/papers/sat21_procount.pdf.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: https://arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: https://arxiv.org/abs/1907.05000.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: https://arxiv.org/abs/1809.09319.